

Raft under Physical Communication Models: A Systematic Review of What Relativity Changes

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Abstract. Consensus protocols such as Raft are specified against abstract communication models: asynchronous or partially synchronous message passing, possibly with loss and partitions. Physical systems add constraints those models do not name explicitly: a finite signal speed, clocks that measure proper time, and, in relativistic settings, no observer-independent simultaneity. This review asks which of these constraints are already accounted for by existing results, which reduce to known communication models, and which remain open. We follow PRISMA 2020 [1] and the software-engineering guidelines of Kitchenham and Charters [2], combining database search with backward and forward snowballing [3].
[Results and conclusions after screening.]

1 Introduction

Distributed-systems courses present consensus against an idealised network in which messages are delayed arbitrarily but eventually delivered, and processes share no clock [4], [5]. A physicist reading these models naturally asks whether they survive contact with the physical world. Signals propagate no faster than light, clocks on moving or gravitationally separated nodes accumulate different proper times, and relativity denies a global notion of “now” on which specifications such as linearizability [6] appear to depend. Lamport’s logical clocks were themselves motivated by special relativity [7], yet the question of whether a concrete, widely deployed protocol remains correct under physical constraints has only recently been studied formally [8], [9], [10].

This review takes Raft as its object because it is widely deployed, precisely specified, and separates cleanly into a timing-independent safety core and timing-dependent mechanisms. The review’s purpose is to establish the boundaries of current knowledge before any new analysis is attempted: what is known, what follows directly from known models, and what is genuinely open.

[One paragraph stating the contributions once results exist.]

2 Scope and boundaries

2.1 Object of study

The object is Raft and leader-based crash-fault-tolerant state-machine replication whose results transfer to Raft (Multi-Paxos, Viewstamped Replication). Byzantine protocols are excluded except where they provide the only analysis of a communication model.

2.2 Communication models

We consider a hierarchy of models ordered by the constraints they impose: synchronous; partially synchronous [5]; asynchronous [4]; asynchronous with loss, omission and partial partitions; delay- and disruption-tolerant networks; and relativistic spacetime, in which message delivery is constrained by the causal (light-cone) structure and each process measures its own proper time. Clock-based mechanisms (leases, bounded-drift assumptions, failure detectors) are included wherever a model’s timing assumptions enter a correctness argument.

2.3 Out of scope

Time-transfer engineering without a computational model, quantum communication, blockchain papers that use Raft only as a black-box ordering service, and performance tuning that does not state a communication model.

3 Research questions

The review addresses four questions. The first two map the field; the third isolates the physically grounded literature; the fourth identifies the gap from which the research questions of the subsequent analytical phase will be drawn.

RQ1 (mapping) Under which communication models have Raft and equivalent protocols been analysed, for which properties (safety, liveness, latency, availability), and with which methods (proof, model checking, analytical model, simulation, testbed)?

RQ2 (timing) Which Raft mechanisms depend on timing or clock assumptions, whether for safety or only for liveness, and how are those assumptions stated?

RQ3 (physics) Which models of distributed computation in relativistic spacetime exist, and which correctness results do they establish for consensus and replicated state machines?

RQ4 (gap) To what extent have clock-dependent mechanisms been analysed under relativistic or otherwise physically grounded models of time?

4 Review method

4.1 Protocol

The protocol was fixed before screening and follows PRISMA 2020 [1] for reporting and Kitchenham and Charters [2] for conduct. It is maintained alongside this article (`review/protocol.md`); deviations are reported in Section 7.

4.2 Sources and search strings

Scopus and Web of Science are the primary databases, complemented by IEEE Xplore, the ACM Digital Library, dblp, and arXiv (cs.DC) for recent preprints. Search strings combine a protocol block (Raft, state-machine replication) with either a communication-model block or a physics-and-time block. The acronym “RAFT” collides with a polymerisation technique and an optical-flow network, so both are excluded explicitly. Full strings and narrowing rules are given in the protocol.

4.3 Snowballing

Database search is complemented by snowballing [3]. Backward snowballing covers the reference lists of all included studies. Forward snowballing starts from a seed set chosen because each seed is either foundational to the relativistic strand or central to Raft’s behaviour under non-ideal networks, while having a citation count small enough to screen completely. [List seed set and iteration counts.]

4.4 Eligibility and screening

[Criteria summary; single-reviewer screening with blinded re-screen of a 20% sample and Cohen’s κ .]

4.5 Data extraction

For each included study we extract the communication model, the properties analysed, the method, the timing mechanisms relied upon, any physical parameters, and the principal result.

5 Results

[PRISMA flow diagram (figure). Mapping table for RQ1. Timing-dependence table for RQ2. Synthesis for RQ3 and RQ4.]

6 Discussion

[Answers to RQ1–RQ4. Explicitly separate (i) results established in the literature, (ii) consequences that follow directly from known models, and (iii) open problems; the latter define the analytical phase.]

7 Threats to validity

[Single reviewer; recency bias toward 2025–2026 preprints; known-item seeding of the pilot search; database coverage of workshop venues.]

8 Conclusion

[Summary and research questions for the analytical phase.]

References

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